

Report 239

Symmetric Sign Matrices and Free Degree Constraints

Leading asymptotics, a critical window, and controlled inverses

5 October 2026

Abstract

Let A_n count symmetric $n \times n$ matrices with entries in $\{-1, +1\}$ and nonnegative row sums, with diagonal entries included. This is OEIS A027832. An exact graph encoding adds one distinguished vertex whose degree is unrestricted. We prove that removing a fixed number f of degree caps near half the graph order multiplies the all-capped graph count by $\Phi(\ell)^{-f}(1 + o(1))$, where $\phi(\ell) = \ell\Phi(\ell)$. More generally, uniformly for $f = O(\sqrt{M})$ free vertices, a finite-saddle formula and a fully explicit equivalent retain a collective quadratic correction and a bounded-offset linear correction. The argument combines uniform dense degree-sequence enumeration with relative graph concentration, an exact exponential tilt, exponential-square uniform integrability, and explicit balancing of the even degree-sum lattice. Applying the established all-capped asymptotic gives a leading relative equivalent for A_n with an essential factor $\exp(\ell\sqrt{n})$ on even orders. We derive the precision of a continuous inverse and rigorous eventual two-ceiling threshold bounds. Exact finite checks and reproducible TeX and Python accompany the proof. The result is a leading $1 + o(1)$ theorem; no all-orders count expansion, effective onset, or historical priority is asserted.

[write] **Status and trust boundary, added 6 October 2026, batch 108.** This is bundle Report 239 of an external AI-assisted research session, filed in ProveIt as the single-source research report **a027832-symmetric-sign-matrices** (provenance, the sources as the write read them, relation to the repository, reading conventions and the collected non-claims in Section 1.2). It is unrefereed and not formalized; its author line and PDF author field read “Report 239”. *What rests on what.* The proof imports three published results, cited and not reproved: McKay–Wormald’s dense degree-sequence enumeration (Theorem 3.1), and McKay–Wanless–Wormald’s all-capped count (their Corollary 2, here (8.5)) and finite-saddle estimates (their Lemma 2 and equation (29), here (8.3) and (8.2)). The constants q , Z_0 , J , the exponential rate and the square-root boundary effect of Theorem 2.2 are therefore McKay–Wanless–Wormald’s; what the source proves is the transfer from all-capped to mixed capped and free coordinates (Theorem 2.1), which turns their count into the matrix equivalent through the graph bridge credited to Greenhill–McKay. The write read the statements of the McKay–Wanless–Wormald results used and the Greenhill–McKay correspondence, and found them as the source quotes them; it did not read the 1990 McKay–Wormald paper. No proof uses a floating-point computation. The write also restored two characters lost in the delivered Section 7 (the subscript “even” of two parity indicators, which printed as “meven”), and added Remark 1.2 (the OEIS entry and the historical table), Remark 9.3 (Section 9 against the transseries volume, with a proof that A_n is strictly increasing for $n \geq 1$) and Section 12 (open questions).

Contents

1	The matrix model and its graph encoding	3
2	Leading equivalents and the critical window	7
3	The uniform degree sequence input	8
4	Relative concentration for mixed constraints	9
5	A uniform sub Gaussian bound	11

6	The exact saddle and the product model	13
7	Entropy weights and parity balancing	14
8	The explicit offset correction and matrix constants	17
9	Controlled inversion of the leading model	19
10	Exact computations and reproducible outputs	22
11	Scope and further questions	23
12	Further questions and research	24
	References	25

1 The matrix model and its graph encoding

A symmetric sign matrix is a symmetric matrix $X = (x_{ij})_{1 \leq i, j \leq n}$ with every $x_{ij} \in \{-1, +1\}$. Its diagonal is variable. Let A_n be the number for which every row sum is nonnegative. Symmetry makes the column conditions identical. Positions are labelled: simultaneous permutations of rows and columns are not identified. We set $A_0 = 1$ by the empty-matrix convention; the published sequence begins with $A_1 = 1$ and $A_2 = 5$ [1].

For $M \geq 1$, $0 \leq k \leq M - 1$, and $0 \leq f \leq M$, let $U_f(M, k)$ count simple graphs on the fixed vertex set $[M]$ such that the first $M - f$ degrees are at most k . The last f vertices are specified in advance and unrestricted. Write $B(M, k) = U_0(M, k)$. There is no factor $\binom{M}{f}$ in this definition.

Proposition 1.1 (Exact graph bridge). *For every $n \geq 0$,*

$$A_n = U_1(n + 1, \lfloor n/2 \rfloor). \quad (1.1)$$

For $n \geq 1$, one also has the exact soft-boundary formula

$$A_n = 2^n \sum_{\substack{G \text{ on } [n] \\ \Delta(G) \leq k}} 2^{-N_k(G)}, \quad k = \lfloor n/2 \rfloor, \quad (1.2)$$

where $N_k(G)$ is the number of vertices of degree k .

Proof. Add a distinguished vertex $*$ to $[n]$. Put an edge ij , $i \neq j$, precisely when $x_{ij} = -1$, and an edge $i*$ precisely when $x_{ii} = -1$. Each of the $n(n + 1)/2 = \binom{n+1}{2}$ independent signs specifies one possible edge, so the correspondence is bijective. The number of negative entries in row i is exactly d_i , giving row sum $n - 2d_i$. Hence the row condition is $d_i \leq \lfloor n/2 \rfloor$, with no condition at $*$.

Alternatively, fix only the negative off-diagonal entries, giving a graph G on $[n]$. If $d_i > k$, no diagonal sign is allowed. If $d_i = k$, the diagonal must be positive. If $d_i < k$, either diagonal sign is allowed. Multiplication over vertices gives $2^{n - N_k(G)}$, proving (1.2). $\square$

The diagonal-to-extra-vertex correspondence is a prior structural fact: Greenhill and McKay state it explicitly for symmetric 0–1 matrices with diagonal weight one [4, p. 902; preprint p. 3]. Proposition 1.1 includes the short proof to fix the sign convention and the inequality cap. Their dense Theorem 1.4 treats prescribed exact row sums with a variable diagonal, a direct pointwise predecessor; summing to the capped event still requires the relative estimates developed here.

The extra vertex is useful because it replaces the boundary weight in (1.2) by a single free degree coordinate. Treating the row sums as independent would lose the relevant rare-event correlation. A multivariate central limit theorem with an absolute error is likewise insufficient: the desired probability decays exponentially in n .

1.1 What is imported and what is proved here

McKay and Wormald’s uniform dense degree-sequence enumeration [2, Theorem 3(ii)] is the pointwise counting input. McKay, Wanless, and Wormald [3] already determined the all-capped graph equivalent, including its exponential rate, constant, and square-root boundary effect. We use their Corollary 2 for the final matrix constants, and their Lemma 2 and equation (29) for the first correction to an auxiliary saddle mass.

The proof below supplies the passage from all-capped to mixed capped and free coordinates, including the window $f = O(\sqrt{M})$. It does not apply the comparison theorem [3, Theorem 3] directly to 2^{-N_k} . That theorem requires a polynomial bound on the ratio of an observable’s maximum to its comparison-model expectation; this condition has not been established for the weight in (1.2). Instead, the proof sums the pointwise enumeration formula with an exact tilt and controls its positive exponential weight. The switching argument is an unequal-cap adaptation of the method in [3, Lemma 4], and its counts are derived explicitly here.

Flammenkamp’s university page identifies a joint counting project with Sillke in 1981 and links an exact table agreeing with A_n through order 16 [5]. This dates the project, not necessarily every value in the current table. The underlying unpublished manuscript or proof notes cited by OEIS were not recovered. Anderson’s cited Section 3.1 [6] was also not inspected. Greene and Kleitman’s complete displayed article [7] was read; it concerns symmetric chains, strengthened Sperner bounds, lattice paths, and two-row plane partitions, and does not state the sign-matrix asymptotic used here. These are precise limits on the inspected historical record, not evidence that no earlier unpublished result exists.

Later neighboring work addresses loopy exact margins, partial stability at exponential-rate scale, and friendly bisections at constant-factor precision [4, 8, 9]. The inspected results do not directly supply the mixed-cap relative equivalent below. In particular, broad partial-constraint ideas and the graph bridge are not presented as new. The present article makes no priority claim.

Remark 1.2 ([write] The OEIS entry and the historical table, 6 October 2026). The live entry A027832 [1] (revision #6 of 12 November 2018, the revision the source inspected; read 6 October 2026) is named, verbatim, “Number of symmetric $\{-1, +1\}$ matrices of order n with nonnegative row and column sums.” Its offset is 1, its 17 data terms $A_1, \dots, A_{17}$ equal the source’s transcription (`data/oeis_prefix.json`), and the package’s exact recursion of Section 10 regenerates all of them. Besides the data the entry has an example ($A_2 = 5$, the five matrices listed), two references (Anderson’s *Combinatorics of Finite Sets*, Ch. 3.1, and “Torsten Sillke and Achim Flammenkamp, unpublished”) and one link (Greene–Kleitman); it has no formula, no program and no asymptotic statement. Theorem 2.2 is therefore the only asymptotic for A027832 in the entry or in the repository. Nothing in the entry is corrected, and nothing was submitted to the OEIS.

The historical table. The write fetched the table the source cites [5] on 6 October 2026. It is headed “number of symmetric 0-1-matrices with each row- and columnsum $\geq n/2$ ” and lists $n = 1, \dots, 16$; its 16 values equal $A_1, \dots, A_{16}$. The two counts are the same: under $+1 \mapsto 1, -1 \mapsto 0$ a row with j entries $+1$ has sign sum $2j - n$, which is nonnegative exactly when $j \geq n/2$, and the diagonal entry is counted once in each. This confirms the source’s statement that the table agrees with A_n through order 16.

The leading formula against the data. The batch-108 intake evaluated A_n divided by the right side of (2.6) for all 17 terms (40-digit `mpmath`, ℓ found to 40 digits; rerun by the write; the delivered noncertified receipt `data/diagnostic_receipt.json` agrees to the digits shown):

n	1	2	3	4	5	6	7	8	9
ratio	0.85922	1.01920	1.00947	1.00152	0.99615	1.00043	0.99616	1.00005	0.99682
n	10	11	12	13	14	15	16	17	
ratio	0.99985	0.99733	0.99974	0.99771	0.99967	0.99800	0.99963	0.99822	

These are the intake’s numbers: 0.9996 at $n = 16$ and 0.998 at $n = 17$, which the intake read as tending to 1 on both parities. In detail, the odd-order ratios increase from $n = 5$ on; the even-order

ratios lie within $4.4 \cdot 10^{-4}$ of 1 from $n = 6$ on, cross 1 between $n = 8$ and $n = 10$, and from $n = 8$ to $n = 16$ move slowly away from 1. Seventeen terms neither confirm nor contradict an $o(1)$: Theorem 2.2 gives the limit 1 along each parity, with no rate and no sign (Question 3).

1.2 Provenance, sources and reading conventions

[write] **Added 6 October 2026, batch 108.** *Provenance.* This report is Report 239 of the session bundle that ProveIt’s docs/incoming intake processed as batch 108: the archive `Report239.zip` (567,302 bytes, SHA-256 `01d2635f9e90...4d80c1d9`, 31 files in a wrapper directory `Report239/`), arrival commit `60f54ea06`, placed in `602e5bd0f`. The text is the delivered `article.tex` with its twelve files `sections/*.tex` (792 lines in all, dated 5 October 2026), shipped under the same names. The delivered 19-page PDF and the checksum manifest `MANIFEST.sha256` (30 entries, verified at placement) are not shipped, and the delivered README is replaced by the report README; all three are retrievable from the arrival commit. The package records no ProveIt commit and cites nothing in the repository. The phrase “private review” occurs in it only in sentences saying that such material is not included (Section 10 and `SOURCES.md`; the delivered README likewise excludes “private audit files”); the PDF author field is “Report 239”. It is a single-source report, so no merge choice was made. The write prefixed the 74 delivered labels with `ssm:` and updated the 67 references to them; it added this subsection, the notes marked [write], Remarks 1.2 and 9.3 and Section 12. One delivered defect was repaired: `sections/07_transfer.tex` contained two carriage-return bytes where the two characters `\r` of `\rm even` had been converted, so the delivered PDF printed “meven” in (7.10) and in the unnumbered display before the final limit of Section 7; both now read `\rm even` (note there). Apart from this repair and the label prefixes, no theorem, proof or number of the source was changed: the added statements are the last of their sections, the added section comes after the last numbered one, and the added displays are unnumbered, so every theorem, section and equation keeps its delivered number.

The sources, as the write read them (6 October 2026).

- OEIS A027832, revision #6, and Flammenkamp’s table (Remark 1.2).
- McKay–Wanless–Wormald [3], the authors’ final version (`deghalfn_final.pdf`): Corollary 2 (p. 3), Lemma 2 and the parametrization (1) (p. 6), equations (17)–(18) (pp. 11–12), (19)–(26) (pp. 12–13) and (27)–(30) (p. 14). Checked against the source: Corollary 2 counts graphs on n vertices with maximum degree at most $k = \frac{1}{2}n - 1 + \tau$, $\tau = o(n^{1/4})$, as $2^{\binom{n}{2}} \zeta_0 \zeta_1^n \exp(\zeta_2 \tau n^{1/2} + \zeta_3 \tau^2 + \zeta_4 \tau^3 n^{-1/2} + o(1))$ with $\zeta_0 = Z_0$, $\zeta_1 = e^{-\ell^2}/(\ell\sqrt{2\pi}) = q$, $\zeta_2 = 2\ell$, $\zeta_3 = J$ and ζ_4 the cubic coefficient quoted after (8.5), so (8.5) is its case $n = M$, $\tau = \theta_M = O(1)$. Lemma 2 gives $S(r, k) = \Phi(L + T) + O(n^{-1})\phi(L + T)$ for bounded L, T , with $r = \frac{1}{2} - \frac{1}{2}Ln^{-1/2}$ and $k = \frac{1}{2}n - 1 + \frac{1}{2}Tn^{1/2}$, which is (8.3); equation (29) gives $L/L_0 = 1 - O(n^{-1})$ at every extremum in their interval I_0 , which is (8.2); their $\mu(r) = Nr^2/(r^2 + (1 - r)^2)$ of (26) is $Hp(r)$; their $\alpha(r)$ of (22) at $n = M$ equals $\frac{M}{2}\mu_M(r)$, half the mean of the capped sum (the write checked this from (22), the exponents $\frac{n}{2} \pm \frac{1}{2}Tn^{1/2} = k + 1$, $n - 1 - k$, and $\mathbb{P}\{\text{Bin}(n - 1, r) \leq k\} = \mathbb{P}\{\text{Bin}(n - 2, r) \leq k - 1\} + (1 - r)\mathbb{P}\{\text{Bin}(n - 2, r) = k\}$), so their saddle equation $\alpha(r) = \mu(r)$ is (6.1); and their (17) states the comparison (3.2) uniformly over concentrated sequences, with γ_2 of (18) equal to γ of (3.1). Not read: their Theorem 3, Lemma 4 and equation (14).
- Greenhill–McKay [4], arXiv:1103.0080v3: the abstract and the paragraph on p. 3 of the preprint that maps a loop to an edge to one new vertex (for row sums with diagonal weight one), as the source says. Their Theorem 1.4 was not read.
- The transseries volume `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex`: `p0:def:core`, `p0:thm:lambert-core`, `p0:prop:factorial-core`, `p0:thm:core-reversion`, `p0:def:three-inverses` and `p0:thm:staircase`, read for Remark 9.3.
- Not read by the write: McKay–Wormald [2] (the hypotheses and uniformity of Theorem 3.1 as stated are the source’s reading of their Theorems 2 and 3(ii)); Anderson [6] (the source did not read it either);

Greene–Kleitman [7]; Dandi–Gamarnik–Zdeborová [8]; Minzer–Sah–Sawhney [9]; and the Liebenau–Wormald paper named in `SOURCES.md`. What is said about them is the source’s.

What was checked at intake. The batch-108 dossier read the manuscript and found no error, and reran the package on a copy (Windows, Python 3.14.4): `build.py` with `--verify-only` verified 30 files; `exact_counts.py` and `constant_certificate.py` reproduced the delivered receipts byte for byte after line-ending conversion; `diagnostics.py` reproduced its receipt except in the last one or two digits of three binary64 values (the ratios `exact_over_S_based_model` at $M = 13$, $f = 1, 3, 6$, for example ...151 against ...156), which the receipt itself labels noncertified floating-point output of the platform’s `libm`; `guard_tests.py` stops on Windows at its own path guard; the PDF and ZIP rebuilds were not run. It also evaluated the leading formula against all 17 terms (table in Remark 1.2). The write rechecked by hand the bridge, the identities $q^{n+1}/\Phi(\ell) = q^n e^{-\ell^2/2}$ and $a^2/(1-2v) = \ell^2 \tau^2/[2(1+2\ell^2)]$, the switching gain (4.5), the tail bound (5.3) and the coefficients of (9.6); it checked the change of variables of Remark 9.3 and the reversion coefficients in SymPy, and compared the historical table. Finite computations are observations, not certificates.

Relation to the repository. Before batch 108 no file of the repository mentioned A027832, symmetric sign matrices or graphs with degrees capped near half the order. No statement of this report is formalized in Lean or Rocq, and its place in the collection confers no formal status. The neighbouring reports are listed in the report README; none needs a reciprocal note.

Reading conventions (letters the manuscript reuses, with the tempting false readings; no symbol of the source was renamed; symbols of the transseries volume carry the subscript “vol” in Remark 9.3).

Symbol	Meaning here (where)	Not to be read as
ℓ	the root of $\phi(\ell) = \ell\Phi(\ell)$, 0.50605... (2.1)	a logarithm; L_M , $L_0(T)$
Φ , ϕ	standard normal distribution function and density	the volume’s core Φ_0^{vol}
q , q_j , q_x	the base $\Phi(\ell)e^{-\ell^2/2}$ (McKay–Wanless–Wormald’s ζ_1); binomial masses (Lemma 5.1); capped masses (Section 7)	each other
v	the variance constant $(1 - 2\ell^2)/4$	v_{vol} of <code>p0:prop:factorial-core</code>
S_M	the cap probability at the finite saddle; McKay–Wanless–Wormald’s $S(r, k)$ at $r = r_M$, $n = M$	a sum
θ_M , τ	cap offset $k - M/2 + 1$ (McKay–Wanless–Wormald’s τ); $\tau = \lim f/\sqrt{M}$	each other
h , h_M	$\phi(x) - x\Phi(x)$ (Section 2); cap probability (Section 4)	each other; h_{vol}
f	number of free vertices	f_{vol} of <code>p0:thm:core-reversion</code>
$N_k(G)$, $N_M(\mathbf{d})$, $N_s(x)$	vertices of degree k ; graphs with degrees $\mathbf{d}$; parity threshold index	each other; $\min(N_0, N_1)$, not N_s , is the staircase N_*^{vol}
$B(M, k)$, B	all-capped count; a coefficient in the proof of Proposition 9.1	McKay–Wanless–Wormald’s $B = \binom{n-1}{k}$
a , b	in Section 9: $\frac{1}{2} \log 2$, $\frac{1}{2} \log 2 + \log q$; elsewhere degrees, vertices, endpoints, a_d , a_M	each other; a_{vol} , b_{vol} of <code>p0:thm:lambert-core</code>
t , u	continuous index and $\sqrt{y/a}$ (Section 9); $t = M^{1/2+\delta}$ (Section 4); a real argument (Lemma 5.1)	t_{vol} , u_{vol}
r_M , r_s	saddle edge probability; continuous inverse	each other
D , Δ	cap $\lceil M/2 + t \rceil$, coefficient D of (9.6), relative entropy $D(p p_0)$; Δ_M mean deficit, $\Delta(G)$ maximum degree	each other
W_M	entropy weight (7.3)	a Lambert function (none occurs)

[write] **What is not claimed, collected from the source.** Both theorems are leading $1 + o(1)$ statements: no rate for the $o(1)$, no higher count coefficient, no all-orders expansion and no numerically certified starting index is asserted, and the auxiliary estimate (8.4) does not strengthen the qualitative error of the enumeration.

The all-capped equivalent, its constants and its square-root boundary effect are McKay–Wanless–Wormald’s; the diagonal-to-extra-vertex correspondence is Greenhill–McKay’s; “broad partial-constraint ideas and the graph bridge are not presented as new”, and the article “makes no priority claim”. The comparison theorem of [3, Theorem 3] is not applied to the weight 2^{-N_k} , because its hypothesis was not established for it. Nothing is proved for $f/\sqrt{M} \rightarrow \infty$, for nonuniform caps beyond bounded offsets, or for deleted or fixed diagonals, which give different counts. The inverse statements give no certified finite-input rounding rule; Proposition 9.2 holds only beyond an ineffective threshold, and more digits of ℓ cannot replace the missing error bound. The historical search was bounded: Anderson’s chapter and the unpublished Silke–Flammenkamp work were not inspected, and “These are precise limits on the inspected historical record, not evidence that no earlier unpublished result exists”. The constant certificate encloses constants only, the diagnostics are noncertified, the manifest establishes “integrity, not authorship or independent provenance”, the guards are “not a sandbox”, and reproduction is claimed only for the recorded toolchain.

2 Leading equivalents and the critical window

Let

$$\phi(x) = \frac{e^{-x^2/2}}{\sqrt{2\pi}}, \quad \Phi(x) = \int_{-\infty}^x \phi(t) dt,$$

and let ℓ be the unique positive solution

$$\phi(\ell) = \ell\Phi(\ell). \quad (2.1)$$

To see existence and uniqueness directly, $h(x) = \phi(x) - x\Phi(x)$ has $h(0) > 0$ and $h'(x) = -\Phi(x) - 2x\phi(x) < 0$ for $x \geq 0$, while $h(1) < 0$. In fact $0 < \ell < 1/\sqrt{2}$: at $x = 1/\sqrt{2}$, the bound $\Phi(x) > 1/2 + x\phi(x)$ gives $x\Phi(x) > \phi(x)$ because $x > \phi(x)$. Define

$$\begin{aligned} q &= \Phi(\ell)e^{-\ell^2/2}, & Z_0 &= \frac{\exp(5\ell^2/4 - \ell^4/2)}{\sqrt{1 + 2\ell^2}}, \\ J &= -\frac{4\ell^2}{1 + 2\ell^2}, & v &= \frac{1 - 2\ell^2}{4} \in (0, 1/4). \end{aligned} \quad (2.2)$$

Theorem 2.1 (Mixed degree constraints). *Fix finite constants $C, K \geq 0$. Let $M \rightarrow \infty$, let k be integral with $|k - M/2| \leq K$, and let $0 \leq f \leq C\sqrt{M}$ be integral. Choose the finite saddle $r_M = 1/2 - L_M/(2\sqrt{M})$ described in Section 6, and put*

$$S_M = \mathbb{P}\{\text{Bin}(M - 1, r_M) \leq k\}, \quad \theta_M = k - M/2 + 1.$$

Then, uniformly over these k and f ,

$$\frac{U_f(M, k)}{B(M, k)} = (1 + o(1))S_M^{-f} \exp \left\{ \frac{\ell^2}{2(1 + 2\ell^2)} \frac{f^2}{M} \right\}. \quad (2.3)$$

Equivalently, with no implicit saddle mass,

$$\frac{U_f(M, k)}{B(M, k)} = (1 + o(1))\Phi(\ell)^{-f} \exp \left\{ -\frac{2\theta_M \ell}{1 + 2\ell^2} \frac{f}{\sqrt{M}} + \frac{\ell^2}{2(1 + 2\ell^2)} \frac{f^2}{M} \right\}. \quad (2.4)$$

In particular, $U_f/B \sim \Phi(\ell)^{-f}$ if $f = o(\sqrt{M})$, including every fixed f .

The equivalence of (2.3) and (2.4) requires the source-based estimate

$$\log S_M = \log \Phi(\ell) + \frac{2\theta_M \ell}{1 + 2\ell^2} M^{-1/2} + O(M^{-1}), \quad (2.5)$$

proved in Section 8. The mere convergence $S_M \rightarrow \Phi(\ell)$ would not suffice when f grows like $\sqrt{M}$. If $f/\sqrt{M} \rightarrow \tau$, the quadratic term has the nonvanishing limit $\ell^2 \tau^2 / [2(1 + 2\ell^2)]$. The linear term remembers the bounded offset of the cap. Both matter on this scale.

Theorem 2.2 (Symmetric sign matrices). *With the constants in (2.2),*

$$A_n \sim 2^{n(n+1)/2} q^n Z_0 e^{-\ell^2/2} \begin{cases} \exp(\ell\sqrt{n} + J/4), & n \text{ even}, \\ 1, & n \text{ odd}. \end{cases} \quad (2.6)$$

Each equivalence is a relative $1 + o(1)$ statement along its parity subsequence.

Useful decimal values, displayed for orientation, are

$$\begin{aligned} \ell &\approx 0.50605446898918076324, & \Phi(\ell) &\approx 0.69359079375820639821, \\ q &\approx 0.61023039919762778828, & Z_0 &\approx 1.08387834752412285047, \\ J &\approx -0.67740809782986162079, & \Phi(\ell)^{-1} &\approx 1.44177230868582047454. \end{aligned}$$

Ordinary high-precision decimal evaluations and any exact interval certificates in the computational package are distinguished there. The decimal display itself is not used to establish the theorem.

The diagonal convention is essential. It is responsible for the extra free vertex in (1.1); deleting or fixing the diagonal produces a different count. The parity factor is also essential: omitting it from even orders would not yield a relative equivalent. Neither theorem asserts a rate for its $o(1)$, higher asymptotic count coefficients, or a numerically certified starting index.

3 The uniform degree sequence input

Put $H = \binom{M}{2}$. For an integer vector $\mathbf{d} = (d_1, \dots, d_M)$ with even sum $2m$, define

$$p = \frac{m}{H}, \quad \bar{d} = \frac{2m}{M}, \quad \gamma(\mathbf{d}) = \frac{1}{(M-1)^2} \sum_{i=1}^M (d_i - \bar{d})^2. \quad (3.1)$$

Let $N_M(\mathbf{d})$ be the number of labelled simple graphs with exactly these degrees.

Theorem 3.1 (Imported dense enumeration). *There exists a sufficiently small positive ε_* such that, uniformly for graphical even-sum degree vectors satisfying*

$$\max_i |d_i - \bar{d}| = O(M^{1/2+\varepsilon_*}), \quad \min(\bar{d}, M-1-\bar{d}) > cM/\log M$$

for a fixed $c > 2/3$, one has

$$N_M(\mathbf{d}) = (1 + o(1)) g(\gamma(\mathbf{d}), p) p^m (1-p)^{H-m} \prod_{i=1}^M \binom{M-1}{d_i}, \quad (3.2)$$

where

$$g(x, p) = \sqrt{2} \exp \left\{ \frac{1}{4} - \frac{x^2}{4p^2(1-p)^2} \right\}. \quad (3.3)$$

This is [2, Theorem 3(ii)], with uniformity supplied by its underlying Theorem 2; it is restated probabilistically in [3, equations (17)–(18)]. To translate that restatement, the graph-model probability of $\mathbf{d}$ at edge probability p is $N_M(\mathbf{d})p^m(1-p)^{H-m}$, whereas the independent-binomial probability is $\prod_i \binom{M-1}{d_i} p^{2m}(1-p)^{2H-2m}$. Dividing gives (3.2). In particular, the exponent contains γ^2 , although γ is already a scaled second moment. We use this theorem only at relative order $1 + o(1)$.

Choose once and for all $\delta > 0$ so small that $3\delta < \varepsilon_*$ and $\delta < 1/16$, and set

$$\mathcal{C} = \left\{ \mathbf{d} : \max_i |d_i - M/2| \leq M^{1/2+2\delta} \right\}. \quad (3.4)$$

On this window, $p = 1/2 + O(M^{-1/2+2\delta})$ and $\max_i |d_i - \bar{d}| \leq 2M^{1/2+2\delta} \leq M^{1/2+3\delta}$ eventually. The density and concentration hypotheses therefore hold uniformly.

Lemma 3.2 (No nongraphical mass in the window). *For all sufficiently large M , every integer vector in $\mathcal{C}$ with even sum is graphical.*

Proof. Sort the degrees in decreasing order and let $a = \min_i d_i$ and $b = \max_i d_i$. Both equal $M/2 + o(M)$, and every entry lies in $[0, M-1]$. By the Erdős–Gallai criterion, it remains to check

$$\sum_{i=1}^j d_i \leq j(j-1) + \sum_{i=j+1}^M \min(j, d_i), \quad 1 \leq j \leq M.$$

For $j \leq a$, the right side is $j(M-1)$, at least the left side. For $j \geq a$, its excess over the left side is at least

$$j(j-1) + (M-j)a - jb = Ma + j^2 - j(a+b+1) \geq Ma - \frac{(a+b+1)^2}{4} = \frac{M^2}{4} + o(M^2) > 0.$$

The even-sum condition completes the criterion. $\square$

We must still justify discarding degree vectors outside $\mathcal{C}$ relative to the small graph probability of interest. That is a statement about the actual mixed graph model, not just the later independent comparison model. The next section supplies it before any use of (3.2) in a sum.

4 Relative concentration for mixed constraints

Take a random graph with independent edge probability $1/2$. Let E_f be the event that its first $M-f$ degrees are at most $k = M/2 + O(1)$. Then

$$\mathbb{P}(E_f) = 2^{-H} U_f(M, k), \quad \mathbb{P}(E_0) = 2^{-H} B(M, k).$$

We prove the following stronger uniformity than is needed for the critical window.

Lemma 4.1. *Uniformly over $0 \leq f \leq M$ and bounded offsets $k - M/2$,*

$$\mathbb{P}(\mathbf{d} \in \mathcal{C} \mid E_f) = 1 - o(1). \quad (4.1)$$

Proof. We first recall the product-measure correlation inequality used here. Two increasing functions of independent Bernoulli coordinates have nonnegative covariance. An induction on the number of coordinates proves it: condition on the last coordinate, use the induction hypothesis for the conditional covariance, and observe that both conditional means are increasing functions of the remaining Bernoulli variable. The covariance of two increasing functions of one Bernoulli variable is nonnegative. The same statement follows for two decreasing functions by negation; opposite monotonicities give nonpositive covariance. This is the Harris–FKG inequality in the form required below.

Every cap event is decreasing in the edges. If

$$h_M = \mathbb{P}\{\text{Bin}(M-1, 1/2) \leq k\},$$

then $h_M \rightarrow 1/2$ uniformly for bounded offsets. Repeated positive association gives

$$\mathbb{P}(E_f) \geq \mathbb{P}(E_0) \geq h_M^M \geq 3^{-M} \quad (4.2)$$

for large M . This rough exponential lower bound is enough to control very small unconditional edge-count tails.

Write $t = M^{1/2+\delta}$ and let m denote the number of edges. A binomial Chernoff bound, or the exponential moment proof of Hoeffding’s inequality, gives

$$\mathbb{P}(m < H/2 - Mt \mid E_f) \leq 3^M \exp\{-cM^{1+2\delta}\} = o(1) \quad (4.3)$$

for an absolute $c > 0$. Indeed $m \sim \text{Bin}(H, 1/2)$ and the squared deviation divided by H is of order $M^{1+2\delta}$. Also, a free vertex’s event $d_i > M/2 + t$ is increasing, while E_f is decreasing. Hence

$$\mathbb{P}(d_i > M/2 + t \mid E_f) \leq \mathbb{P}\{\text{Bin}(M-1, 1/2) > M/2 + t\} \leq e^{-c'M^{2\delta}}.$$

A union bound over at most M vertices is still $o(1)$.

It remains to rule out low degrees after these two exceptional events have been removed. Fix $m \geq H/2 - Mt$. Impose cap k on constrained vertices and cap $D = \lceil M/2 + t \rceil$ on free vertices. Call these caps c_i ; all are at most D , while every cap is at least $k = M/2 + O(1)$ for sufficiently large M . The conditional law, given m and these caps, is uniform over its allowed graphs.

Fix a vertex v , and let a_d be the number of allowed graphs with degree d at v . For $d < \min_i c_i$, consider a switch that deletes an edge ab and inserts av , where $a \notin N(v) \cup \{v\}$. The vertex a keeps its degree, b loses one, and v gains one, so all caps are respected. Counting oriented edges by their first endpoint, the number of forward switches is exactly

$$2m - d - \sum_{a \in N(v)} d_a \geq 2m - d(D+1) =: L_d.$$

This count already excludes $b = v$, because a is not adjacent to v .

In a target graph with degree $d+1$ at v , a reverse switch first chooses $a \in N(v)$ and then a nonneighbor b of a other than a . Ignoring a possible cap violation at b gives an upper bound

$$(d+1)(M-1) - \sum_{a \in N(v)} d_a.$$

The vertices outside $N(v) \cup \{v\}$ number $M - d - 2$ and have total degree at most $(M - d - 2)D$. Therefore

$$\sum_{a \in N(v)} d_a \geq 2m - (d+1) - (M - d - 2)D,$$

so the number of reverse switches is at most

$$R_d = (d+1)M - 2m + (M-d-2)D.$$

Switches with their ordered endpoints reverse uniquely. Double counting yields

$$a_d L_d \leq a_{d+1} R_d. \quad (4.4)$$

The gain is visible without a hidden asymptotic cancellation:

$$L_d - R_d = 4m - (M+1)d - M - (M-2)D.$$

Using $d \leq M/2 - 10t$, $m \geq H/2 - Mt$, and $D \leq M/2 + t + 1$ gives

$$L_d - R_d \geq 5Mt + 12t - \frac{5}{2}M + 2 \geq c_1 Mt \quad (4.5)$$

for large M . Also $R_d = O(M^2)$ whenever the target class is nonempty. If a lower class is nonempty, $L_d > 0$ forces a nonempty next class; a zero reverse bound cannot occur in that case. Thus (4.4) implies

$$a_{d+1} \geq (1 + c_2 M^{-1/2+\delta}) a_d \quad (d \leq M/2 - 10t)$$

through every relevant nonempty class.

Put $a = \lfloor M/2 - M^{1/2+2\delta} \rfloor$ and $b = \lfloor M/2 - 10t \rfloor$. Then $b - a \sim M^{1/2+2\delta}$ and b lies below every cap. For each $d \leq a$, iteration to level b gives $a_d \leq a_b \exp(-c_3 M^{3\delta})$; if all such lower classes are empty the conclusion is immediate. Summing at most M possible lower degrees and dividing by the total number of allowed graphs gives

$$\mathbb{P}(d_v \leq a \mid m, \text{caps}) \leq M e^{-c_3 M^{3\delta}}.$$

This is uniform in v, m, f . A union bound over v , followed by restoration of the discarded events in (4.3) and the free upper tails, proves (4.1). $\square$

For comparison, positive association also gives the exact elementary bounds

$$1 \leq \frac{U_f(M, k)}{B(M, k)} \leq h_M^{-f}. \quad (4.6)$$

Indeed, conditioned on E_f , the probability that every free vertex also satisfies its cap is at least the product of their unconditional cap probabilities. For $f = 1$, the upper bound tends to 2. These bounds are useful finite checks, but they do not replace the relative concentration argument above.

5 A uniform sub Gaussian bound

The eventual summation weight grows like the exponential of a positive square. Weak convergence alone cannot justify its expectation. We first establish a uniform bound strong enough to do so.

Lemma 5.1 (Capped binomial exponential moments). *Let X have the distribution of $\text{Bin}(b, r)$ conditioned on $X \leq k$, where $0 < r < 1$ and $0 \leq k \leq b$ are arbitrary. Then*

$$\text{Var}(X) \leq \frac{b}{4}, \quad (5.1)$$

and, for every real u ,

$$\mathbb{E} e^{u(X - \mathbb{E}X)} \leq e^{bu^2/8}. \quad (5.2)$$

The same statements hold without conditioning.

Proof. Write the binomial as the sum of b independent Bernoulli variables and condition their sum to be at most k . The conditional coordinates are exchangeable. For $b \geq 2$, let $F(j) = \mathbb{P}\{\text{Bin}(b-2, r) \leq j\}$, extended by zero below its support and by one above it. The unnormalized probabilities for two selected coordinates are

$$(1-r)^2 F(k), \quad r(1-r)F(k-1), \quad r(1-r)F(k-1), \quad r^2 F(k-2).$$

Their covariance has the sign of $F(k-2)F(k) - F(k-1)^2$.

To check this sign directly, let q_j be the binomial mass sequence, extended by zero. Its successive positive ratios decrease. Consequently, for $i \leq j$, $q_{j+1}q_{i-1} \leq q_j q_i$. Sum over $i \leq j$ to obtain $q_{j+1}F(j-1) \leq q_j F(j)$. Since

$$F(j)^2 - F(j-1)F(j+1) = q_j F(j) - q_{j+1} F(j-1),$$

F is log-concave, including the support boundaries. Thus the conditional coordinates are pairwise negatively correlated. If a is their common conditional success probability, then

$$\text{Var}(X) \leq ba(1-a) \leq b/4.$$

The cases $b \leq 1$ are immediate.

Exponential tilting by e^{uX} preserves the capped-binomial family, replacing r by

$$r_u = \frac{re^u}{1-r+re^u}.$$

The second derivative of its log moment-generating function is the variance under this tilted law, hence is bounded by $b/4$ for every real u . Integrating twice from zero proves (5.2). Without a cap, the same variance and tilting argument applies. $\square$

Corollary 5.2. *If $X_1, \dots, X_M$ are independent, each ordinary or capped $\text{Bin}(M-1, r)$ variables, set*

$$Z_M = \frac{\sum_i X_i - \sum_i \mathbb{E}X_i}{M}.$$

For $M \geq 2$,

$$\mathbb{P}(|Z_M| > u) \leq 2 \exp \left\{ -\frac{2M}{M-1} u^2 \right\}, \quad u \geq 0. \quad (5.3)$$

For each fixed $\alpha < 2$, $\sup_M \mathbb{E}e^{\alpha Z_M^2} < \infty$. If a_M stays in a bounded set, there exists $\eta > 0$ such that

$$\sup_M \mathbb{E}e^{(1+\eta)(Z_M+a_M)^2} < \infty. \quad (5.4)$$

Proof. Multiplying (5.2) and optimizing the exponential Markov bound gives (5.3). For $0 < \alpha < 2$, integration of that tail gives, for example,

$$\mathbb{E}e^{\alpha Z_M^2} \leq 1 + 4\alpha \int_0^\infty u e^{-(2-\alpha)u^2} du = 1 + \frac{2\alpha}{2-\alpha}.$$

Nonpositive α are immediate. Finally, for any $\rho > 0$,

$$(z+a)^2 \leq (1+\rho)z^2 + (1+\rho^{-1})a^2.$$

Choose small fixed $\rho, \eta > 0$ with $(1+\rho)(1+\eta) < 2$ and use boundedness of a_M . This proves (5.4). $\square$

The gap between the required quadratic coefficient 1 and the admissible coefficient 2 is the strict margin that permits the critical number of free coordinates. It also controls third and fourth centered moments at orders $M^{3/2}$ and M^2 , respectively, by integration of the single-coordinate sub-Gaussian tail.

6 The exact saddle and the product model

For $r \in (0, 1)$, let X_r be $\text{Bin}(M-1, r)$ conditioned to be at most k , and define

$$\mu_M(r) = \mathbb{E}X_r, \quad p(r) = \frac{r^2}{r^2 + (1-r)^2}.$$

We use one saddle for all f , chosen by the exact identity

$$\mu_M(r_M) = (M-1)p(r_M), \quad r_M = \frac{1}{2} - \frac{L_M}{2\sqrt{M}}, \quad 0 \leq L_M \leq 1. \quad (6.1)$$

Lemma 6.1. *For sufficiently large M , a root in (6.1) exists, uniformly for bounded offsets $k - M/2$. Every such root has $L_M \rightarrow \ell$. Moreover,*

$$S_M \rightarrow \Phi(\ell), \quad \frac{\text{Var}(X_{r_M})}{M} \rightarrow v, \quad \frac{\Delta_M}{\sqrt{M}} \rightarrow \frac{\ell}{2}, \quad (6.2)$$

where $\Delta_M = (M-1)r_M - \mu_M(r_M)$.

Proof. Write $b = M-1$ and $r = 1/2 - L/(2\sqrt{M})$ with $0 \leq L \leq 1$. The binomial central limit theorem is uniform on this compact range of parameters, and its centered moments are uniformly integrable by the usual binomial exponential-moment bound. Its standardized upper endpoint is

$$\frac{k - br}{\sqrt{br(1-r)}} = L + O(M^{-1/2}).$$

Writing $\lambda(L) = \phi(L)/\Phi(L)$, the elementary truncated normal identities are

$$\mathbb{E}[Z \mid Z \leq L] = -\lambda(L), \quad \text{Var}(Z \mid Z \leq L) = 1 - L\lambda(L) - \lambda(L)^2,$$

obtained by integration using $\phi'(z) = -z\phi(z)$. Hence

$$\mu_M(r) = br - \frac{1}{2}\sqrt{M}\lambda(L) + o(\sqrt{M}), \quad p(r) = \frac{1}{2} - LM^{-1/2} + O(M^{-3/2}),$$

uniformly. Subtraction gives

$$\frac{\mu_M(r) - (M-1)p(r)}{\sqrt{M}} = \frac{1}{2}\{L - \lambda(L)\} + o(1). \quad (6.3)$$

The limiting function is negative at 0, positive at 1, and has its unique zero at ℓ , by (2.1). Continuity gives a finite- M root, and compactness plus uniform convergence forces every such root to converge to ℓ . Finite- M uniqueness is unnecessary.

The binomial cap probability tends to $\Phi(\ell)$. The truncated variance identity, with $\lambda(\ell) = \ell$, gives the variance limit $v = (1 - 2\ell^2)/4$. Finally the exact saddle yields

$$\Delta_M = (M-1)(r_M - p(r_M)) = \frac{1}{2}\ell\sqrt{M} + o(\sqrt{M}).$$

All assertions are uniform over bounded offsets by the same compactness argument. $\square$

Set $r = r_M$, $p_0 = p(r_M)$, and $m_0 = Hp_0$. Under a product law R_f , take $M - f$ independent copies of X_r and f independent ordinary $\text{Bin}(M - 1, r)$ variables. These are degree coordinates for a comparison calculation, not the degrees of an independent-edge graph. Define

$$Y_M = \frac{\sum_i X_i - 2m_0}{M}, \quad a_M = \mathbb{E}_{R_f} Y_M = \frac{f\Delta_M}{M}. \quad (6.4)$$

The saddle identity ensures that $a_M = 0$ for $f = 0$.

Lemma 6.2. *Along any sequence with bounded $k - M/2$ and $f/\sqrt{M} \rightarrow \tau \in [0, \infty)$,*

$$Y_M \implies \mathcal{N}(a, v), \quad a = \tau\ell/2, \quad \gamma(\mathbf{X}) \longrightarrow v \quad \text{in probability}, \quad (6.5)$$

and $R_f(\mathcal{C}) \rightarrow 1$. The means a_M are uniformly bounded for $f \leq C\sqrt{M}$, and the exponential-square moment bound (5.4) applies to Y_M uniformly in that window.

Proof. The total variance divided by M^2 tends to v : the $M - f$ capped coordinates each contribute $vM + o(M)$, while replacing $f = o(M)$ of them changes the normalized variance by $O(f/M)$. The third absolute centered moments are $O(M^{3/2})$ uniformly by Lemma 5.1, so their sum divided by the total variance to the power $3/2$ is $O(M^{-1/2})$. The triangular-array Lyapunov theorem therefore gives a centered normal limit of variance v . Equation (6.2) gives $a_M \rightarrow a$.

For the sample second moment, center each coordinate at its own mean. Its fourth moment is $O(M^2)$, so the variance of $M^{-2} \sum_i (X_i - \mathbb{E}X_i)^2$ is $O(M^{-1})$. Its mean tends to v . The difference between the two group means is $\Delta_M = O(\sqrt{M})$, whose additional squared contribution divided by M^2 is $O(f\Delta_M^2/M^2) = O(f/M) = o(1)$. Cross terms vanish by Cauchy–Schwarz. Subtracting the square of the sample mean error changes the normalized second moment by $o_{\mathbb{P}}(1)$, and replacing M^2 by $(M - 1)^2$ has no effect. This proves the assertion about γ .

Every coordinate’s mean differs from $M/2$ by $O(\sqrt{M})$. The single-coordinate bound (5.2) therefore gives

$$R_f(\mathcal{C}^c) \leq 2M \exp\{-cM^{4\delta}\} = o(1)$$

for large M . The square-moment statement is Corollary 5.2, since $Y_M = Z_M + a_M$ with a_M bounded. $\square$

7 Entropy weights and parity balancing

We now sum the uniform enumeration formula. This is the step where relative tail control and exponential-square integrability are both needed.

7.1 An exact change of measure

For an even-sum degree vector $\mathbf{d} \in \mathcal{C}$ satisfying the $M - f$ caps, its R_f probability is

$$R_f(\mathbf{d}) = S_M^{-(M-f)} r^{2m} (1-r)^{2H-2m} \prod_i \binom{M-1}{d_i}.$$

Define the positive common factor

$$K_M = 2^{-H} S_M^M p_0^{m_0} (1-p_0)^{H-m_0} r^{-2m_0} (1-r)^{2m_0-2H}. \quad (7.1)$$

Real exponents cause no difficulty; m_0 need not be integral. Division of (3.2) by the product mass gives

$$2^{-H} N_M(\mathbf{d}) = (1 + o(1)) K_M S_M^{-f} g(\gamma(\mathbf{d}), p) W_M(m) R_f(\mathbf{d}), \quad (7.2)$$

uniformly in the window, where the remaining factor is exactly

$$\begin{aligned} W_M(m) &= \exp\{HD(p\|p_0)\}, \\ D(p\|p_0) &= p \log \frac{p}{p_0} + (1-p) \log \frac{1-p}{1-p_0}. \end{aligned} \quad (7.3)$$

For verification, subtract the logarithm of the weight at m_0 . The coefficient of the linear term vanishes because

$$\log \frac{p_0}{1-p_0} = 2 \log \frac{r}{1-r}.$$

The remainder is precisely the binary relative entropy in (7.3). Its sign is positive.

7.2 Uniform integrability of the positive weight

Since $p - p_0 = MY_M/(2H)$ and $p, p_0 \rightarrow 1/2$ uniformly on $\mathcal{C}$, Taylor's theorem gives

$$0 \leq HD(p\|p_0) \leq (1 + o(1)) Y_M^2 \quad \text{on } \mathcal{C}. \quad (7.4)$$

The leading coefficient is $M^2/[8Hp_0(1-p_0)] \rightarrow 1$. On each set $|Y_M| \leq A$ with fixed A , one also has

$$HD(p\|p_0) = Y_M^2 + o(1) \quad (7.5)$$

uniformly. For example, the mean-value form of the entropy Taylor expansion proves (7.4) directly, without estimating a large cubic remainder separately.

Corollary 5.2 and boundedness of a_M imply, after decreasing $\eta > 0$ if necessary,

$$\sup_{\substack{M, f \leq C\sqrt{M} \\ |k-M/2| \leq K}} \mathbb{E}_{R_f} [(\mathbf{1}_{\mathcal{C}} W_M)^{1+\eta}] < \infty \quad (7.6)$$

for all sufficiently large M . Thus these weights are uniformly integrable. The same holds for $e^{Y_M^2}$. Compact approximation using (7.5), $R_f(\mathcal{C}) \rightarrow 1$, and then the two uniform-integrability bounds yields

$$\mathbb{E}_{R_f} \left| \mathbf{1}_{\mathcal{C}} W_M - e^{Y_M^2} \right| \rightarrow 0. \quad (7.7)$$

This is an L^1 replacement, so it remains valid after multiplication by the even-sum indicator.

Furthermore, $0 < g(x, p) \leq \sqrt{2}e^{1/4}$ and Lemma 6.2 gives

$$g(\gamma(\mathbf{X}), p) \rightarrow g_0 := \sqrt{2}e^{1/4-4v^2} \quad \text{in probability on } \mathcal{C}. \quad (7.8)$$

Here g may be defined arbitrarily off $\mathcal{C}$ without affecting a weighted expression supported on $\mathcal{C}$. Boundedness of g and Hölder's inequality with (7.6) imply that replacing g by g_0 introduces $o(1)$ in the weighted expectation.

7.3 The even degree sum contributes one half

For all sufficiently large M , $f \leq C\sqrt{M}$ leaves a capped coordinate. Its mass function q_x , extended by zero to all integers, is unimodal. Since $r \rightarrow 1/2$ and S_M stays bounded away from zero, the standard binomial mode bound gives

$$\max_x q_x = O(M^{-1/2}), \quad \sum_x |q_x - q_{x-1}| = 2 \max_x q_x.$$

Let F be a bounded Lipschitz function and fix the sum T of the other coordinates. Define

$$A_T = \sum_x (-1)^x q_x F((T + x - 2m_0)/M).$$

Shifting the sum by one and averaging bounds it by

$$|A_T| \leq \|F\|_\infty \max_x q_x + \frac{\text{Lip}(F)}{2M} = o(1), \quad (7.9)$$

uniformly in T . Indeed the two contributions are respectively half the total variation of q and half the change in F under a step $1/M$.

Conditioning on the other coordinates, including their parity sign, shows that even and odd total sums receive equal asymptotic expectations for every such $F(Y_M)$. Approximate e^{y^2} by bounded Lipschitz cutoffs and use the uniform integrability above. Equations (7.7) and (7.8) then show, along any sequence with $f/\sqrt{M} \rightarrow \tau$,

$$\begin{aligned} \mathbb{E}_{R_f} \left[\mathbf{1}_{\{\sum_i X_i \text{ even}\}} \mathbf{1}_{\mathcal{C}} g(\gamma, p) W_M \right] \\ \longrightarrow \frac{g_0}{2} \mathbb{E} e^{(Z+a)^2}, \quad Z \sim \mathcal{N}(0, v), \quad a = \tau \ell / 2. \end{aligned} \quad (7.10)$$

No rare-event local central limit theorem is used for this parity step.

[write] **Restored text, 6 October 2026.** In the delivered file the two characters `\rm even`, in (7.10) and in the display of $\mathbb{P}(E_f)$ in the next subsection, had been converted to a carriage-return byte, so the delivered PDF printed “meven” in the subscript of the indicator. The write restored `\rm even` in both places; the indicator is that of the event that $\sum_i X_i$ is even, as the subsection title says. Apart from the label prefixes, nothing else in this section was changed.

7.4 Completion of the transfer proof

An elementary Gaussian integral, by completing the square, gives

$$\mathbb{E} e^{(Z+a)^2} = \frac{1}{\sqrt{1-2v}} \exp\left(\frac{a^2}{1-2v}\right), \quad v < 1/2. \quad (7.11)$$

Summing (7.2) over even vectors in $\mathcal{C}$ is legitimate because its relative error is uniform and all summands are nonnegative. Lemma 3.2 ensures that no nongraphical even vectors enter. Lemma 4.1 then restores all graph vectors at relative error $1 + o(1)$. Thus

$$\mathbb{P}(E_f) = (1 + o(1)) K_M S_M^{-f} \mathbb{E}_{R_f} \left[\mathbf{1}_{\{\sum_i X_i \text{ even}\}} \mathbf{1}_{\mathcal{C}} g(\gamma, p) W_M \right].$$

For $f = 0$, the final expectation tends to the positive number $g_0/[2\sqrt{1-2v}]$. Dividing the f expression by the $f = 0$ expression, and using $1 - 2v = (1 + 2\ell^2)/2$, yields

$$\frac{U_f(M, k)}{B(M, k)} S_M^f \longrightarrow \exp \left\{ \frac{\ell^2 \tau^2}{2(1 + 2\ell^2)} \right\}$$

whenever $f/\sqrt{M} \rightarrow \tau$.

This sequential statement implies the uniform equivalent (2.3). Otherwise a sequence of counterexamples with $f/\sqrt{M} \in [0, C]$ would have a subsequence on which this ratio converges to some τ , contradicting the preceding limit and continuity of its right side. Bounded cap offsets cause no problem, since all preceding estimates are uniform over them. This proves the finite-saddle part of Theorem 2.1. The remaining explicit simplification and the matrix theorem are established next.

8 The explicit offset correction and matrix constants

8.1 Why the finite saddle mass has the required rate

Set

$$\theta_M = k - M/2 + 1, \quad T_M = 2\theta_M/\sqrt{M}.$$

This is the exact parametrization $k = M/2 - 1 + (T_M/2)\sqrt{M}$ used in [3]. Let $L_0(T)$ be the positive solution

$$\phi(L_0(T) + T) = L_0(T)\Phi(L_0(T) + T). \quad (8.1)$$

In particular $L_0(0) = \ell$. In a neighborhood of zero, smoothness follows from the implicit function theorem because the derivative in L does not vanish.

We carefully identify the finite saddle before importing a quantitative estimate. In [3, Section 6], the function denoted $\alpha(r)$ is half the mean of the sum of M capped-binomial coordinates, and the function denoted $\mu(r)$ is $Hp(r)$. Thus their saddle equation $\alpha(r) = \mu(r)$ is precisely (6.1). Our $m_0 = Hp(r_M)$ lies in their interval

$$I_0 = [H/2 - M^{3/2+2\varepsilon}, kM/2]$$

for any sufficiently small fixed $\varepsilon > 0$ and large M : its distance below $H/2$ is $\frac{1}{2}\ell M^{3/2} + o(M^{3/2})$, and its distance below $kM/2$ has the same positive leading order. Hence the stationary-saddle expansion in their equation (29) applies. For bounded θ_M , and consequently bounded T_M , it gives

$$L_M = L_0(T_M) + O(M^{-1}). \quad (8.2)$$

This uses an actual quantitative source expansion, not a rate inferred from the central limit theorem.

Their Lemma 2, in this same parametrization, gives

$$S_M = \Phi(L_M + T_M) + O(M^{-1}) \quad (8.3)$$

uniformly for the bounded parameters at issue. In particular there is no separate $M^{-1/2}$ term beyond the shift in the argument. The -1 in $k = M/2 - 1 + (T_M/2)\sqrt{M}$ is already part of that lattice-sensitive source formula.

For completeness, differentiating (8.1) yields the derivative used in [3, equation (14)]. Write $\xi = 2L_0(T)^2 + L_0(T)T$. Since $\lambda'(x) = -\lambda(x)(x + \lambda(x))$, the equation $L_0(T) = \lambda(L_0(T) + T)$ implies

$$L'_0(T) = -\xi\{1 + L'_0(T)\}, \quad L'_0(0) = -\frac{2\ell^2}{1 + 2\ell^2}.$$

Equations (8.2) and (8.3) now give

$$L_M + T_M = \ell + \frac{T_M}{1 + 2\ell^2} + O(M^{-1}),$$

and $\phi(\ell)/\Phi(\ell) = \ell$ proves

$$\log S_M = \log \Phi(\ell) + \frac{2\theta_M \ell}{1 + 2\ell^2} M^{-1/2} + O(M^{-1}). \quad (8.4)$$

Multiplying its remainder by $f \leq C\sqrt{M}$ gives $O(M^{-1/2}) = o(1)$. Substitution into (2.3) proves (2.4), and also its $f = o(\sqrt{M})$ corollary. The full enumeration still has only a qualitative $o(1)$ relative error; the auxiliary estimate (8.4) does not strengthen that claim.

8.2 The established all constrained count

The required special case of [3, Corollary 2] is

$$B(M, k) = 2^H Z_0 q^M \exp\{2\ell\theta_M \sqrt{M} + J\theta_M^2 + o(1)\}, \quad \theta_M = O(1). \quad (8.5)$$

The source actually permits $\theta_M = o(M^{1/4})$ and includes a cubic term

$$\frac{4\ell(6\ell^2 - 1)}{3(1 + 2\ell^2)^3} \theta_M^3 M^{-1/2},$$

which is $o(1)$ in the bounded-offset case used here. Its exponential base is written there as $e^{-\ell^2}/(\ell\sqrt{2\pi})$; equation (2.1) makes this exactly q . The constants q, Z_0, J and the square-root boundary effect in (8.5) are therefore attributed to that source.

For the matrix bridge, $M = n + 1$ and $k = \lfloor n/2 \rfloor$. Hence

$$\theta_M = \begin{cases} 1/2, & n \text{ even,} \\ 0, & n \text{ odd.} \end{cases}$$

Theorem 2.1 at $f = 1$ gives $A_n \sim B(n + 1, \lfloor n/2 \rfloor)/\Phi(\ell)$. Moreover

$$\frac{q^{n+1}}{\Phi(\ell)} = q^n e^{-\ell^2/2}, \quad \sqrt{n+1} - \sqrt{n} = O(n^{-1/2}).$$

Substitution into (8.5) proves Theorem 2.2 with exactly the factor shown in (2.6).

The free-vertex critical correction and the matrix parity effect have related origins but different roles. The former comes from a bounded nonzero mean shift in a Gaussian quadratic expectation; the latter is inherited from the lattice-sensitive all-capped count. For one free vertex the critical corrections tend to one, but the matrix's even-order factor $e^{\ell\sqrt{n}}$ remains.

9 Controlled inversion of the leading model

A relative $1 + o(1)$ count equivalent is an $o(1)$ logarithmic approximation. Because the logarithm grows quadratically, this already gives a sharper continuous-index statement than an $o(1)$ index error, but it does not justify a certified finite-input rounding rule.

Let $s = 1$ designate even positive indices and $s = 0$ designate odd positive indices. Put

$$a = \frac{\log 2}{2}, \quad b = \frac{\log 2}{2} + \log q, \quad c_s = \log Z_0 - \frac{\ell^2}{2} + \frac{sJ}{4}, \quad (9.1)$$

and define, for $t > 0$,

$$F_s(t) = at^2 + bt + s\ell\sqrt{t} + c_s. \quad (9.2)$$

Theorem 2.2 is equivalent to

$$\log A_n = F_s(n) + o(1) \quad (9.3)$$

on the indicated parity. For sufficiently large y , let $r_s(y)$ be the unique sufficiently large solution of $F_s(r_s(y)) = y$.

Proposition 9.1 (Continuous inverse and its precision). *The inverse is well-defined eventually and*

$$r'_s(y) \sim \frac{1}{(\log 2)r_s(y)}. \quad (9.4)$$

At the actual sequence values,

$$r_s(\log A_n) = n + o(n^{-1}). \quad (9.5)$$

Writing $u = \sqrt{y/a}$, its elementary model expansion is

$$r_s(y) = u - \frac{b}{2a} - \frac{s\ell}{2a}u^{-1/2} + \left(\frac{b^2}{8a^2} - \frac{c_s}{2a}\right)u^{-1} + O(u^{-3/2}). \quad (9.6)$$

If $y = \log A_n$, the right side without its displayed remainder equals $n + o(u^{-1})$.

Proof. We have

$$F'_s(t) = 2at + b + \frac{s\ell}{2\sqrt{t}} \sim (\log 2)t > 0$$

eventually, and $F_s(t) \rightarrow \infty$. This proves existence, uniqueness on the increasing tail, and (9.4). Equation (9.3) first implies $r_s(\log A_n) \sim n$. The mean value theorem then divides the $o(1)$ logarithmic discrepancy by a derivative asymptotic to $(\log 2)n$, proving (9.5).

For (9.6), substitute $t = u + A + Bu^{-1/2} + Du^{-1}$ in $F_s(t) - y$. Cancellation of the coefficients of u , $u^{1/2}$, and 1 gives

$$A = -\frac{b}{2a}, \quad B = -\frac{s\ell}{2a}, \quad D = \frac{b^2}{8a^2} - \frac{c_s}{2a}.$$

The residual is $O(u^{-1/2})$ and $F'_s(t) \asymp u$, so one further mean value estimate gives the $O(u^{-3/2})$ model remainder. At an actual sequence value, (9.5) contributes only a known $o(u^{-1})$ error, which dominates the finer model remainder. This establishes the last assertion without postulating an unknown count coefficient. $\square$

The parity displacement in this continuous inverse first appears at order $u^{-1/2}$. It is a consequence of the known square-root term in the logarithm; it is not an additional term of a proved all-orders expansion for A_n .

9.1 Integer thresholds need two ceilings

Define

$$\lceil t \rceil_s = 2 \left\lceil \frac{t - (1 - s)}{2} \right\rceil + (1 - s),$$

the least integer at least t of the parity designated by s . Let $N_s(x)$ be the least positive index of that parity with $A_n \geq x$.

Proposition 9.2 (Eventual threshold brackets). *For every $\varepsilon > 0$, once x is sufficiently large,*

$$\lceil r_s(\log x - \varepsilon) \rceil_s \leq N_s(x) \leq \lceil r_s(\log x + \varepsilon) \rceil_s. \quad (9.7)$$

The least unrestricted positive index with $A_n \geq x$ is $\min(N_0(x), N_1(x))$. The required lower bound on x is not made effective here.

Proof. The sequence is nondecreasing: append an all-positive last row and column to a valid matrix. This is an injection, since every existing row sum increases by one and the new row sum is positive. The sequence is unbounded by Theorem 2.2.

For all sufficiently large indices of the indicated parity, (9.3) implies $|\log A_n - F_s(n)| \leq \varepsilon$. Below the lower ceiling in (9.7), an integer of that parity is strictly less than $r_s(\log x - \varepsilon)$, so $\log A_n < \log x$. At the upper ceiling, $F_s(n) \geq \log x + \varepsilon$, hence $\log A_n \geq \log x$. Taking x large enough also excludes the finitely many earlier indices. This proves both bounds. $\square$

The two ceilings may coincide, but the theorem does not certify when they do for a prescribed finite input. Such a certificate requires either an explicit remainder bound with a known validity range or exact counts at adjacent candidate indices. More decimal digits in ℓ or a smaller numerical residual in $F_s(t) = y$ cannot supply that missing asymptotic error bound.

Remark 9.3 ([write] This section against the transseries volume, 6 October 2026). The volume is `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex`; its symbols carry the subscript “vol” here.

(1) *Strict monotonicity* (proved by the write). $A_0 = A_1 = 1$, and $A_{n+1} > A_n$ for every $n \geq 1$.

Proof. The proof of Proposition 9.2 maps each counted matrix of order n injectively to a counted matrix of order $n + 1$ whose last row and column are all $+1$. For $n \geq 1$, the symmetric matrix of order $n + 1$ whose only entry -1 is at position $(n + 1, n + 1)$ has row sums $n + 1$ in its first n rows and $n - 1 \geq 0$ in the last, so it is counted by A_{n+1} but lies outside the image. At $n = 0$ the only matrix of order 1 with nonnegative row sum is $(+1)$, so $A_1 = 1 = A_0$. $\square$

Hence $(A_n)_{n \geq 1}$ is strictly increasing, and the least unrestricted index $\min(N_0(x), N_1(x))$ of Proposition 9.2 is *exactly* the integer staircase $N_*^{\text{vol}}(x)$ of `p0:def:three-inverses(1)` with $n_1 = 1$ (an instance). The parity ceiling $\lceil t \rceil_s$ is exactly the residue-class ceiling $\rho + r[(\nu - \rho)/r]$ of `p0:eq:residue-staircase`, with $r_{\text{vol}} = 2$, $\rho = 1 - s$ and t in place of ν .

(2) *An admissible core, but not a Lambert one.* Put $X_* = \max\{1, |b|/\log 2\}$. On (X_*, ∞) the function F_s is C^2 , $F'_s(t) = t \log 2 + b + s\ell/(2\sqrt{t}) > |b| + b \geq 0$, $F''_s(t) = \log 2 - s\ell/(4t^{3/2}) \geq \log 2 - \ell/4 > 0$ (as $\ell < 1/\sqrt{2}$), and $F'_s(t) \rightarrow +\infty$, $tF'_s(t) \rightarrow +\infty$, $F_s(t) \rightarrow +\infty$. So F_s is an admissible core in the sense of `p0:def:core`, with $\Lambda_{\infty}^{\text{vol}} = +\infty$; for $y > F_s(X_*)$ its core solution is the unique solution of $F_s(t) = y$ in (X_*, ∞) , and for all large y it is $r_s(y)$ (an instance of that definition). The phase is neither of the volume’s two cores, $a_{\text{vol}}X + b_{\text{vol}} \log X$ (`p0:thm:lambert-core`) and $X(\kappa_{\text{vol}} \log X + d_{\text{vol}})$ (`p0:prop:factorial-core`): it has no logarithm, its leading balance is quadratic, and no Lambert function occurs.

(3) *An instance after a change of variables.* The four-term model expansion (9.6) is the start of the unique formal solution in `p0:thm:core-reversion`. Put $t_{\text{vol}} = u^{-1/2}$ and $r_s(y) = u(1 + E)$, so that $y = au^2$. Dividing $F_s(u(1 + E)) - au^2 = 0$ by au^2 gives exactly

$$2E + E^2 + f_{\text{vol}}(t_{\text{vol}}, E) = 0, \quad f_{\text{vol}}(t, w) = \frac{b}{a}t^2(1 + w) + \frac{s\ell}{a}t^3(1 + w)^{1/2} + \frac{c_s}{a}t^4,$$

that is `p0:eq:core-master` with $\Lambda_{\text{vol}} = 2$, $h_{\text{vol}}(w) = w^2$ and $R_{\text{vol}} = \mathbb{Q}[a^{-1}, b, s\ell, c_s]$ (the binomial series of $(1 + w)^{1/2}$ has rational coefficients, and $f_{\text{vol}} \in tR_{\text{vol}}[[t, w]]$). Its unique solution $E = \sum_{n \geq 1} e_n^{\text{vol}} t_{\text{vol}}^n$ has $e_1^{\text{vol}} = 0$, $e_2^{\text{vol}} = -b/(2a)$, $e_3^{\text{vol}} = -s\ell/(2a)$, $e_4^{\text{vol}} = b^2/(8a^2) - c_s/(2a)$, the coefficients A, B, D of the proof of Proposition 9.1, so $u(1 + e_2^{\text{vol}}u^{-1} + e_3^{\text{vol}}u^{-3/2} + e_4^{\text{vol}}u^{-2})$ is the right side of (9.6) without its remainder (checked by the write in SymPy from the triangular recurrence `p0:eq:core-triangular`, the displayed equation re-derived from (9.2)). The volume's theorem is formal: the analytic remainder $O(u^{-3/2})$, and the statement at $y = \log A_n$, are the source's mean-value arguments.

(4) *Analogues only.* Proposition 9.2 is not an instance of `p0:thm:staircase`: r_s is not the inverse of an interpolation of (A_n) (Proposition 9.1 gives only $r_s(\log A_n) = n + o(n^{-1})$, not the identity $\nu_{\text{vol}}(A_n) = n$ of an interpolated inverse), and the brackets come from the two envelopes $F_s \pm \varepsilon$ at the integers of one parity. Its conclusion has the shape of the separation condition `p0:thm:staircase(2)` (when the two ceilings of (9.7) coincide, $N_s(x)$ is their common value) and of the residue-class formula `p0:thm:staircase(4)` with $r_{\text{vol}} = 2$. Likewise (9.5) gives $n = \lfloor r_s(\log A_n) + \frac{1}{2} \rfloor$ for all sufficiently large n of parity s , the analogue of `p0:thm:staircase(3)`, with an ineffective threshold. No novelty is claimed for the inversions.

[write] **Independent check of the write, 6 October 2026.** An adversarial check made by the intake after the write (commit b5026035f) re-derived the statements marked [write] with its own code, after fetching again the OEIS entry A027832, Flammenkamp's table, McKay–Wanless–Wormald's final version and Greenhill–McKay's arXiv:1103.0080v3. *Remark 1.2.* Revision #6 (12 November 2018), its name, offset, example, two references and one link are as quoted, with no formula, program or asymptotic; the table's heading is verbatim and its 16 values equal $A_1, \dots, A_{16}$; literal enumeration gives $A_1, \dots, A_5$. With $\ell = 0.50605446898918\dots$ recomputed at 50 digits, all 17 ratios of the table were reproduced to the digits shown, and so were the statements about them (odd ratios increasing and below 1 from $n = 5$; even ratios within $4.34 \cdot 10^{-4}$ of 1 from $n = 6$, above 1 at $n = 8$, below from $n = 10$, their distance from 1 increasing from $n = 8$ to 16). *Section 1.2.* Against the rendered pages, Corollary 2 (p. 3; $\zeta_0, \dots, \zeta_4 = 1.083878, 0.610230, 1.012108, -0.677408, 0.104696$, recomputed, with $\zeta_1 = q$), Lemma 2 and (1) (p. 6), (17)–(18) (pp. 11–12, uniform over concentrated sequences), (22), (26) and (29) (pp. 12–14) are as the write reads them, and $\alpha(r) = \frac{M}{2}\mu_M(r)$ holds identically in exact rational arithmetic ($6 \leq M \leq 15$, four caps near $M/2$, three values of r); Greenhill–McKay's loop-to-vertex bijection is on p. 3 of the preprint. The restored text: the delivered file has carriage returns at byte offsets 4435 and 5322, each inside `\rm even`. *Remark 9.3.* (1) The injection and the extra matrix were re-checked by hand and against the data; (2) F_s satisfies every clause of `p0:def:core` on (X_*, ∞) , $X_* = 1$ here ($|b|/\log 2 = 0.2126$, $\log 2 - \ell/4 = 0.5666$); (3) the master equation is exact and the triangular recurrence gives $e_1, \dots, e_4 = 0$, A, B, D (SymPy); (4) is accurate. For the record, the remark does not name `plt:thm:lw-template`: with $Z = \sqrt{t}$ and $H = (F_s/a)^{1/4} = Z + \sum_{m \geq 1} \beta_m Z^{-m}$ the model equation $H = \sqrt{u}$ is a degenerate instance of it ($\mu_{\text{vol}} = 0$, no logarithm), whose solution squared reproduces (9.6) through u^{-1} (SymPy); this changes nothing in the remark. *Provenance.* The archive facts (567,302 bytes, the SHA-256 quoted, 31 files, 30 manifest entries, 792 delivered lines, 19 pages) and the counts of 74 labels and 67 references were confirmed. No error was found in the article; the report README printed two roundings as truncations ($Z_0 = 1.08388\dots$, $J = -0.67741\dots$), corrected there with a dated note.

10 Exact computations and reproducible outputs

The accompanying package separates exact combinatorial checks, rational certificates for constants, floating-point diagnostics, and deterministic document reproduction. None of these finite checks replaces the analytic proof or supplies an effective onset for Theorems 2.1 and 2.2.

10.1 An exact labelled graph recursion

For a list of residual caps $\mathbf{c} = (c_1, \dots, c_r)$, let $T(\mathbf{c})$ count graphs on r labelled remaining vertices obeying those caps. A cap above $r - 1$ can be replaced by $r - 1$. Sorting the cap list preserves the count, since relabelling is a bijection. Remove a vertex of cap c_1 , and group the other vertices by their common cap a , with multiplicity m_a . If it chooses j_a neighbors in group a , then

$$T(\mathbf{c}) = \sum_{\substack{0 \leq j_a \leq m_a \ (a \geq 1) \\ \sum_a j_a \leq c_1}} \left(\prod_a \binom{m_a}{j_a} \right) T \left(\text{sort} \bigcup_a \{(a-1)^{[j_a]}, a^{[m_a-j_a]}\} \right). \quad (10.1)$$

Here no neighbor may be selected from cap zero, an exponent in square brackets means multiplicity, and every new cap is truncated to $r - 2$. The base case with at most one vertex is 1. Every neighbor subset is counted once, and $\binom{m_a}{j_a}$ is necessary because the vertices remain labelled even though the memoization key is sorted. Induction on r proves the recurrence.

In particular,

$$B(M, k) = T(k^{[M]}), \quad U_f(M, k) = T(k^{[M-f]}, (M-1)^{[f]}).$$

This computation does not use the asymptotic formulas. It regenerates all 17 published OEIS values $A_1, \dots, A_{17}$ and separately checks the empty convention $A_0 = 1$. For reference, the beginning is

$$1, 5, 14, 315, 2634, 301262, 8035168, 4451407563 \quad (n = 1, \dots, 8).$$

Two direct algorithms supply independent small-case checks. One enumerates all upper-triangular sign assignments, mirrors them, and tests row sums literally. The other enumerates every simple graph edge mask and tests each specified degree cap. The former checks $n = 0, \dots, 5$; the latter checks all mixed uniform-cap/free-coordinate choices at graph orders $M = 0, \dots, 6$. The exact receipt records 113 exhaustive mixed-graph cases, 88 additional multi-free inclusion and product-correlation checks, and 2800 capped-binomial variance inequalities evaluated with rational parameters and integer arithmetic. The all-free value $2^{\binom{M}{2}}$ is also checked. Full ranges and output values are recorded in the receipt.

10.2 Constants and diagnostics have different guarantees

The package contains rational interval certificates for ℓ and the transfer constant. The root is isolated by opposite signs of $\phi(x) - x\Phi(x)$ at rational endpoints and the strict monotonicity proved in Section 2. Rational Taylor bounds and a Machin formula for π enclose the elementary integrals and exponentials used in those sign tests. In particular the verifier establishes

$$\begin{aligned} 0.506054468989180763236198306442 &< \ell < 0.506054468989180763236198306443, \\ 1.441772308685820474543062664826 &< \Phi(\ell)^{-1} < 1.441772308685820474543062664827. \end{aligned}$$

Additional certified constants and the exact arithmetic details are listed in the certificate receipt and computation notes. Certification of these constants is independent of, and does not certify, the asymptotic remainders.

A separate diagnostic script evaluates ratios of exact small counts to the leading model, finite saddle roots, truncated variances, and critical-window correction factors at bounded test parameters, using binary64 arithmetic. Its decimal values are explicitly labelled noncertified. Agreement is useful for catching signs, offsets, and factors of two; it does not prove a convergence rate, a larger free-vertex regime, or an effective finite threshold.

10.3 A frozen and bounded build

The public package includes TeX source, Python programs, frozen receipts, source attribution, and a SHA-256 manifest. It contains no copied third-party paper or private review record. The build verifies source hashes, membership, and directory structure before execution, writes only into a new external output directory, and checks that source bytes remain unchanged afterward.

Public numerical and combinatorial APIs reject invalid or out-of-range inputs before expensive work. These checks use explicit exceptions, not assertions that disappear under Python optimization. Python children are invoked with `-B`; the guard suite contains 170 checks, including hostile inherited settings and optimized execution. Additionally, the integer-to-decimal conversion limit is set to 640 explicitly, independently of the inherited environment. More expensive algorithms have smaller documented bounds. Those bounds are implementation limits and do not restrict the mathematical theorems.

TeX uses a private format and cache, fixed reproducibility metadata, and `-no-shell-escape`. The archive has sorted members, fixed timestamps, stored compression, and fixed regular-file modes. The replay verifier extracts the actual generated ZIP into separate clean trees, rebuilds under ordinary and optimized Python, compares all seven top-level output files byte for byte, checks every archive member and its metadata, and verifies source immutability. Logs are retained separately for inspection. Reproduction is claimed for the recorded Python and TeX toolchain, not arbitrary future versions.

11 Scope and further questions

The proved transfer has two distinct scales. A fixed number of free vertices contributes one factor $\Phi(\ell)^{-1}$ per vertex. At $f \asymp \sqrt{M}$, the cumulative mean shift survives the Gaussian normalization, and the positive quadratic weight produces a collective factor. The bounded offset of the degree cap contributes a linear correction on that same scale. Both are present in (2.4).

Several extensions remain meaningful and require new estimates.

1. **An effective relative remainder.** The present proof uses a uniform $o(1)$ dense enumeration theorem and qualitative limit arguments. An explicit remainder with computable constants would turn the inverse brackets into finite-input certificates. It would need quantitative control of the enumeration input, relative graph tails, tilted moments, and parity error together.
2. **Higher count coefficients.** The computable expansion of the saddle mass is only one ingredient. A next term for A_n requires a correspondingly precise mixed degree-sequence expansion and all associated lattice and moment corrections. Expanding the known leading

model alone does not produce a justified coefficient for the actual count.

3. **More than order $\sqrt{M}$ free vertices.** If $f/\sqrt{M} \rightarrow \infty$, the normalized mean shift a_M is unbounded. The compact Gaussian argument and bounded-shift integrability used here no longer prove (2.3). A moving saddle adapted to the free fraction, together with moderate-deviation estimates, is a natural route; no extrapolated formula is claimed.
4. **Nonuniform caps and marked free degrees.** The graph concentration switching already permits unequal caps. Extending the summation to bounded heterogeneous offsets or generating functions for the free degrees would require choosing and analyzing the corresponding tilt. It may reveal which aggregate offset statistics govern the first collective correction.
5. **Historical identification.** The recovered historical table and direct loopy-graph predecessor narrow the attribution question, but they do not replace inspection of Anderson’s cited chapter or the underlying unpublished Sillke–Flammenkamp work. Further access could clarify the history without changing the proof given here.

The analytic conclusions of this report are therefore the leading relative matrix equivalent, the fixed and bounded-critical free-vertex transfer, and the inverse statements justified by their precision. The exact finite computations support the model and implementation, while the source record identifies the imported ingredients. Neither numerical agreement nor a bounded literature search is a priority certificate.

12 Further questions and research

[write] **Added 6 October 2026, batch 108.** Following Vladimir Reshetnikov’s standing rule of 4 October 2026, every claim of the source that is not proved in full is stated here as an open question, with its source, the argument offered and what is missing; the source’s own five questions are those of Section 11, which stay as printed. The intake and the write found no false statement in the source and narrowed none of its claims; the one delivered defect was typographical (Section 7). What the source does not prove itself is imported and cited, and the questions below record how far the write could check those citations.

1. **The imported enumeration input** (source: Section 3, Theorem 3.1 and the paragraph after it; Section 1.1). The source quotes McKay–Wormald’s Theorem 3(ii), “with uniformity supplied by its underlying Theorem 2”, under the hypotheses $\max_i |d_i - \bar{d}| = O(M^{1/2+\varepsilon_*})$ and $\min(\bar{d}, M - 1 - \bar{d}) > cM/\log M$ with $c > 2/3$, and checks that the window $\mathcal{C}$ satisfies them. *Checked at the write:* the probabilistic restatement in [3, (17)–(18)], uniform over concentrated sequences, and its translation into (3.2) (Section 1.2). *Not checked:* the hypotheses and the uniformity as stated in McKay–Wormald’s 1990 paper, which the write did not read. The other imported statements (Corollary 2, Lemma 2 and equation (29) of [3], and the Greenhill–McKay correspondence) were checked against their texts (Section 1.2).
2. **The historical record** (source: Section 1.1 and Question 5 of Section 11). The Flammenkamp table, dated by its homepage to a 1981 project with Sillke, agrees with $A_1, \dots, A_{16}$ (confirmed by the write, Remark 1.2). *Open:* whether Anderson’s Section 3.1, or the unpublished Sillke–Flammenkamp work cited by the OEIS, states an asymptotic for A_n or for the all-capped count; neither the source nor the write inspected them. The source’s argument is that the inspected record (the OEIS entry, the table, Greene–Kleitman) states none; what is missing is the text of those two items.

3. **The next term and the parity pattern** (added by the write; not a claim of the source; it sharpens the source’s Question 2). Through $n = 17$ the ratio of A_n to the right side of (2.6) (table in Remark 1.2) is below 1 and increasing on odd orders from $n = 5$, and on even orders within $4.4 \cdot 10^{-4}$ of 1 from $n = 6$, below 1 from $n = 10$ and slowly moving away from 1 through $n = 16$. Is there a second term, for instance of relative order $n^{-1/2}$ with a parity-dependent coefficient, and does its sign explain this pattern? The source’s Question 2 explains why expanding the leading model alone cannot answer this; the finite data are observations, not evidence of a rate.

References

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Sections 2–3, particularly Proposition 5, discuss partial stability at an exponential-rate scale. <https://arxiv.org/abs/2305.03591v2>.

- [9] D. Minzer, A. Sah, and M. Sawhney, *On perfectly friendly bisections of random graphs*, Annals of Probability **52** (2024), 2281–2341. Lemma 3.2 and Section 7.2 give the neighboring first-moment comparison at constant-factor precision. doi:10.1214/24-AOP1696. Public full text: <https://par.nsf.gov/servlets/purl/10554553>.